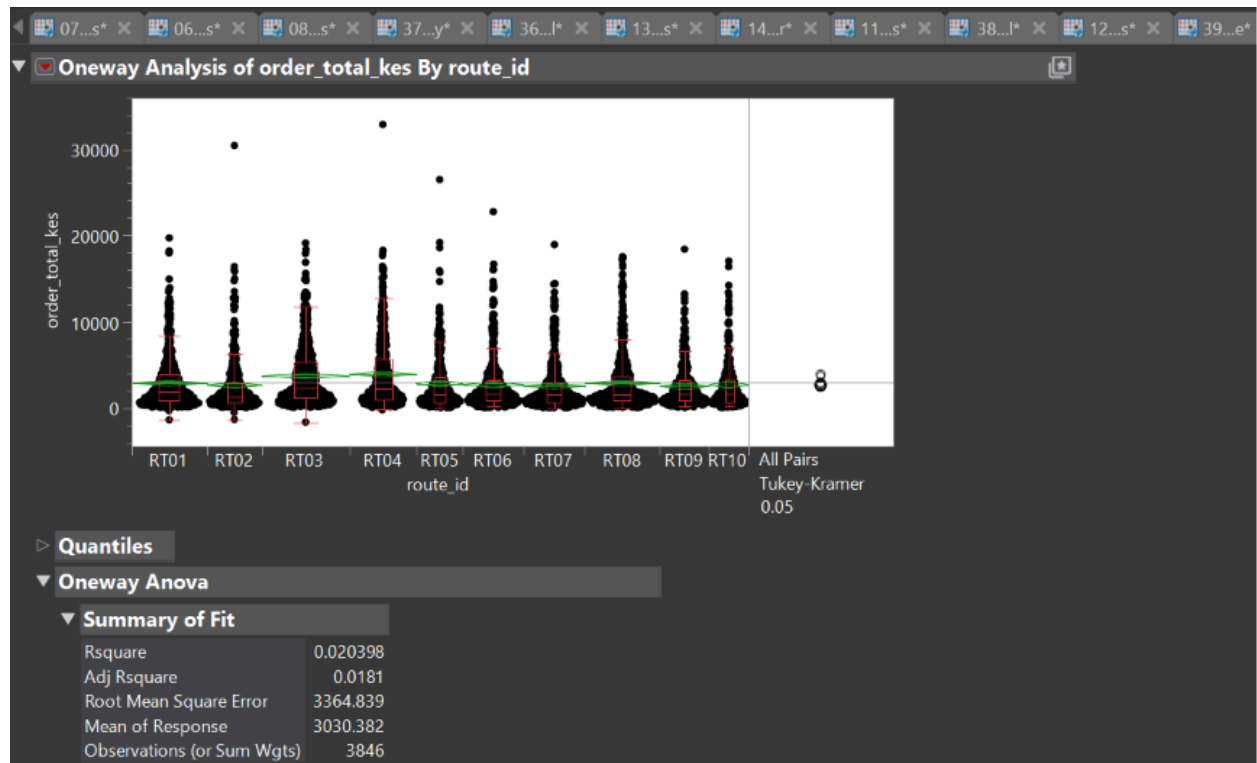


EDA_Bivariate_OrderTotal_by_Route



Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio	Prob > F
route_id	9	904368893	100485433	8.8751	<.0001*
Error	3836	4.3432e+10	11322140		
C. Total	3845	4.4336e+10			

Means for Oneway Anova

Level	Number	Mean	Std Error	Lower 95%	Upper 95%
RT01	471	2913.18	155.04	2609.2	3217.2
RT02	342	2680.75	181.95	2324.0	3037.5
RT03	546	3752.26	144.00	3469.9	4034.6
RT04	417	3952.88	164.78	3629.8	4275.9
RT05	292	2861.41	196.91	2475.3	3247.5
RT06	377	2741.43	173.30	2401.7	3081.2
RT07	385	2576.35	171.49	2240.1	2912.6
RT08	465	2921.93	156.04	2616.0	3227.9
RT09	308	2556.58	191.73	2180.7	2932.5
RT10	243	2723.34	215.85	2300.1	3146.5

Std Error uses a pooled estimate of error variance

Means Comparisons

Comparisons for all pairs using Tukey-Kramer HSD

Confidence Quantile

q*

Alpha

3.16557

0.05

HSD Threshold Matrix

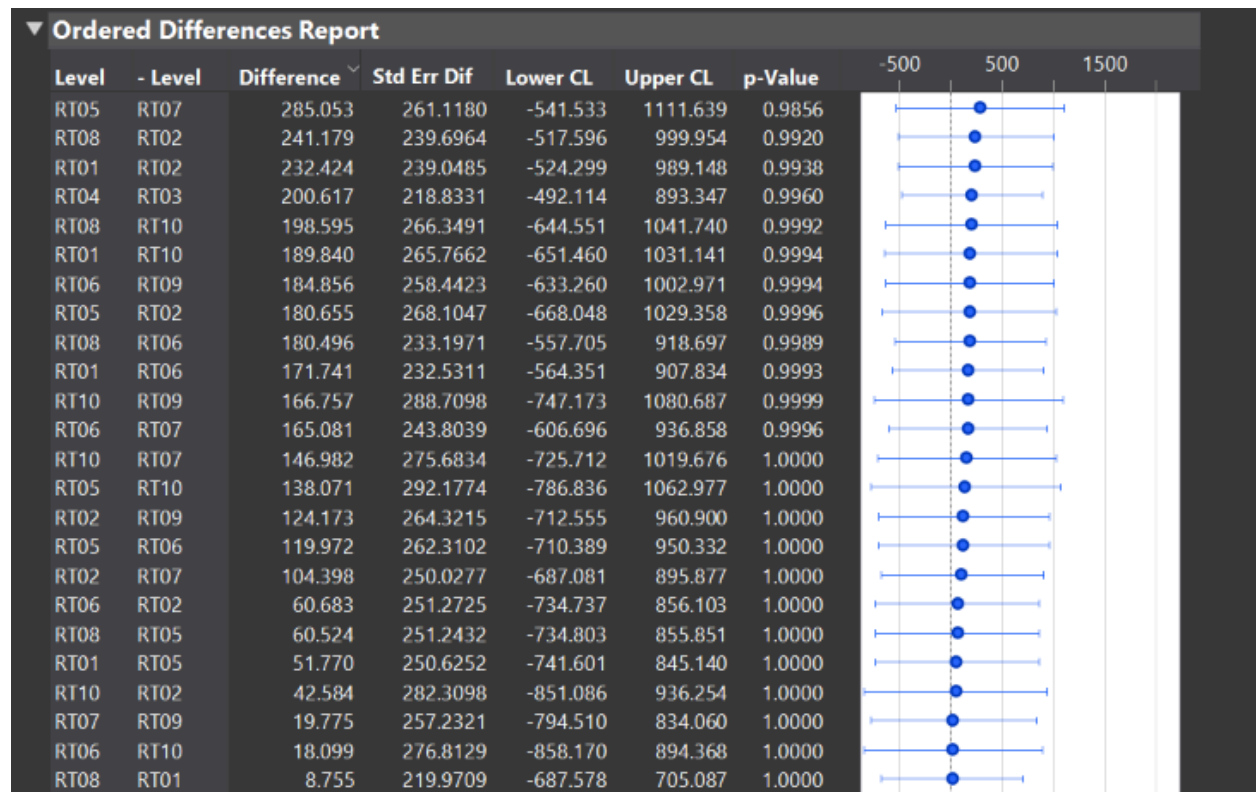
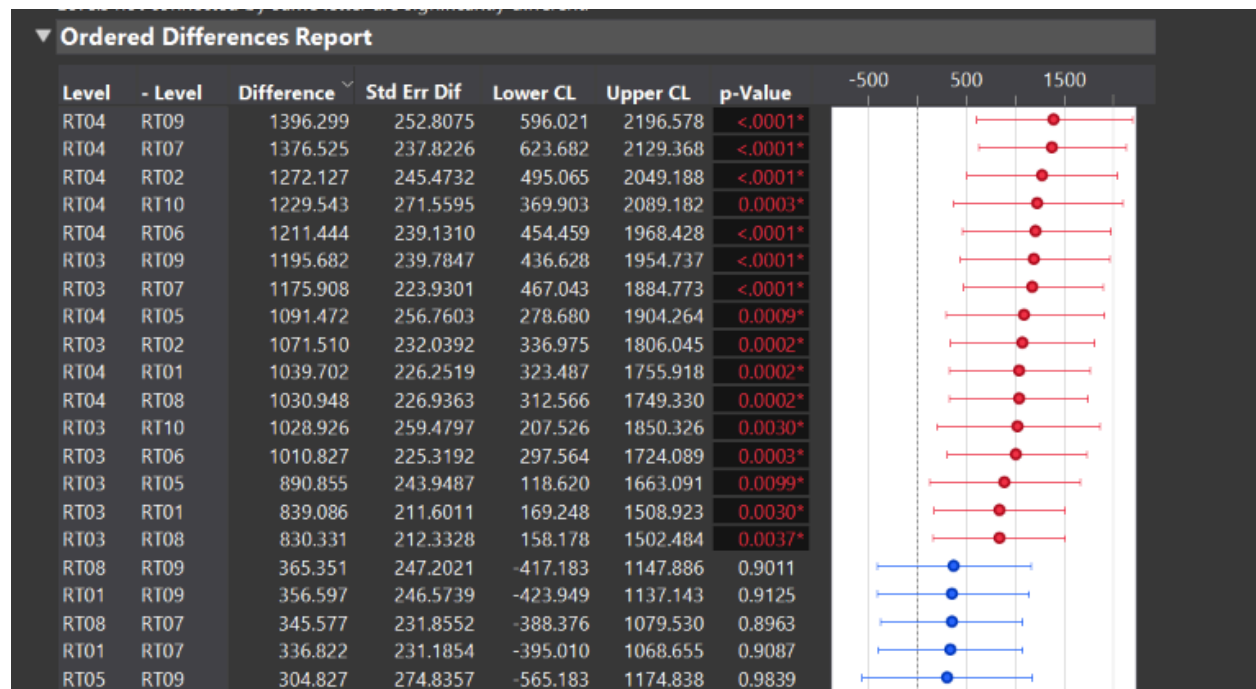
Abs(Dif)-HSD

	RT04	RT03	RT08	RT01	RT05	RT06	RT10	RT02	RT07	RT09
RT04	-737.67	-492.11	312.57	323.49	278.68	454.46	369.90	495.07	623.68	596.02
RT03	-492.11	-644.67	158.18	169.25	118.62	297.56	207.53	336.97	467.04	436.63
RT08	312.57	158.18	-698.56	-687.58	-734.80	-557.70	-644.55	-517.60	-388.38	-417.18
RT01	323.49	169.25	-687.58	-694.10	-741.60	-564.35	-651.46	-524.30	-395.01	-423.95
RT05	278.68	118.62	-734.80	-741.60	-881.53	-710.39	-786.84	-668.05	-541.53	-565.18
RT06	454.46	297.56	-557.70	-564.35	-710.39	-775.82	-858.17	-734.74	-606.70	-633.26
RT10	369.90	207.53	-644.55	-651.46	-786.84	-858.17	-966.33	-851.09	-725.71	-747.17
RT02	495.07	336.97	-517.60	-524.30	-668.05	-734.74	-851.09	-814.55	-687.08	-712.55
RT07	623.68	467.04	-388.38	-395.01	-541.53	-606.70	-725.71	-687.08	-767.72	-794.51
RT09	596.02	436.63	-417.18	-423.95	-565.18	-633.26	-747.17	-712.55	-794.51	-858.33

Positive values show pairs of means that are significantly different.

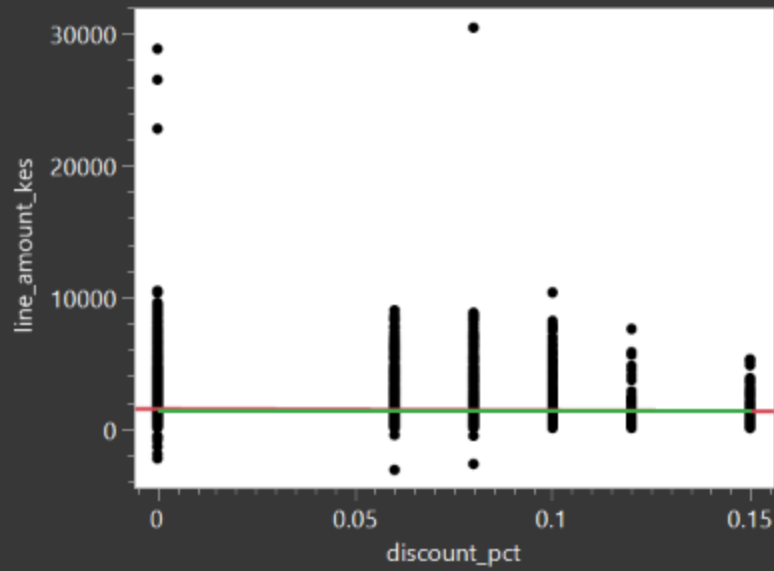
Connecting Letters Report			
Level		Mean	Std Error
RT04	A	3952.9	164.78
RT03	A	3752.3	144.00
RT08	B	2921.9	156.04
RT01	B	2913.2	155.04
RT05	B	2861.4	196.91
RT06	B	2741.4	173.30
RT10	B	2723.3	215.85
RT02	B	2680.8	181.95
RT07	B	2576.4	171.49
RT09	B	2556.6	191.73

Levels not connected by same letter are significantly different.



EDA_Bivariate_LineAmount_vs_Discount.png

▼ **Bivariate Fit of line_amount_kes By discount_pct**



- ☒ Linear Fit
- ☒ Fit Mean

▼ **Linear Fit**

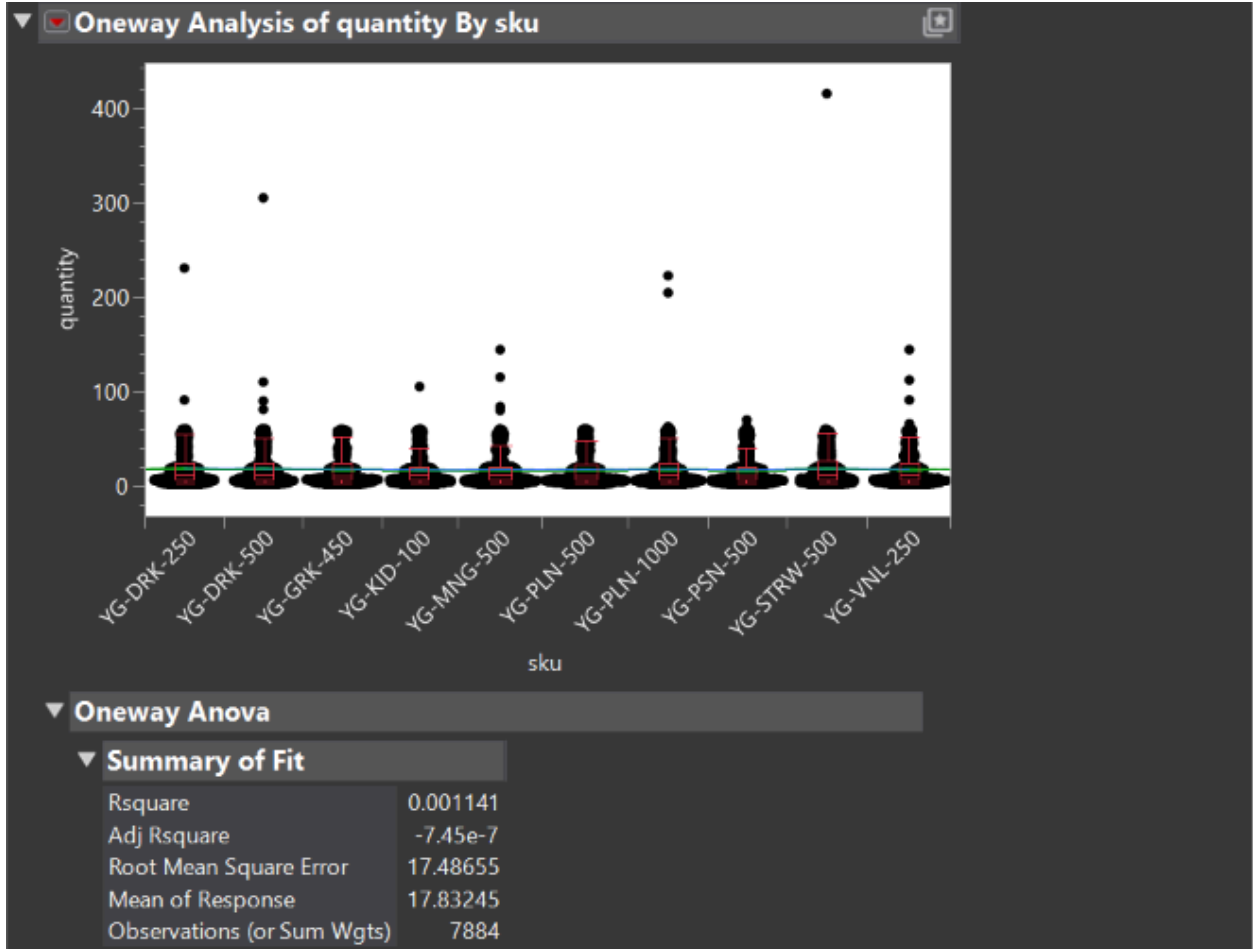
$$\text{line_amount_kes} = 1518.6182 - 1045.842 \cdot \text{discount_pct}$$

▼ **Summary of Fit**

RSquare	0.000766
RSquare Adj	0.000639
Root Mean Square Error	1649.35
Mean of Response	1478.291
Observations (or Sum Wgts)	7884

▼ Lack Of Fit				
Source	DF	Sum of Squares	Mean Square	F Ratio
Lack Of Fit	4	28536074.8	7134019	2.6246
Pure Error	7878	2.1413e+10	2718114	Prob > F
Total Error	7882	2.1442e+10		0.0329*
			Max RSq	0.0021
▼ Analysis of Variance				
Source	DF	Sum of Squares	Mean Square	F Ratio
Model	1	16426722.8	16426723	6.0384
Error	7882	2.1442e+10	2720355.4	Prob > F
C. Total	7883	2.1458e+10		0.0140*
▼ Parameter Estimates				
Term	Estimate	Std Error	t Ratio	Prob> t
Intercept	1518.6182	24.78634	61.27	<.0001*
discount_pct	-1045.842	425.6018	-2.46	0.0140*
▼ Fit Mean				
Mean	1478.291			
Std Dev [RMSE]	1649.877			
Std Error	18.58139			
SSE	2.15e+10			

EDA_Bivariate_Quantity_by_SKU.png



▼ Oneway Anova

▼ Summary of Fit

Rsquare	0.001141
Adj Rsquare	-7.45e-7
Root Mean Square Error	17.48655
Mean of Response	17.83245
Observations (or Sum Wgts)	7884

▼ Analysis of Variance

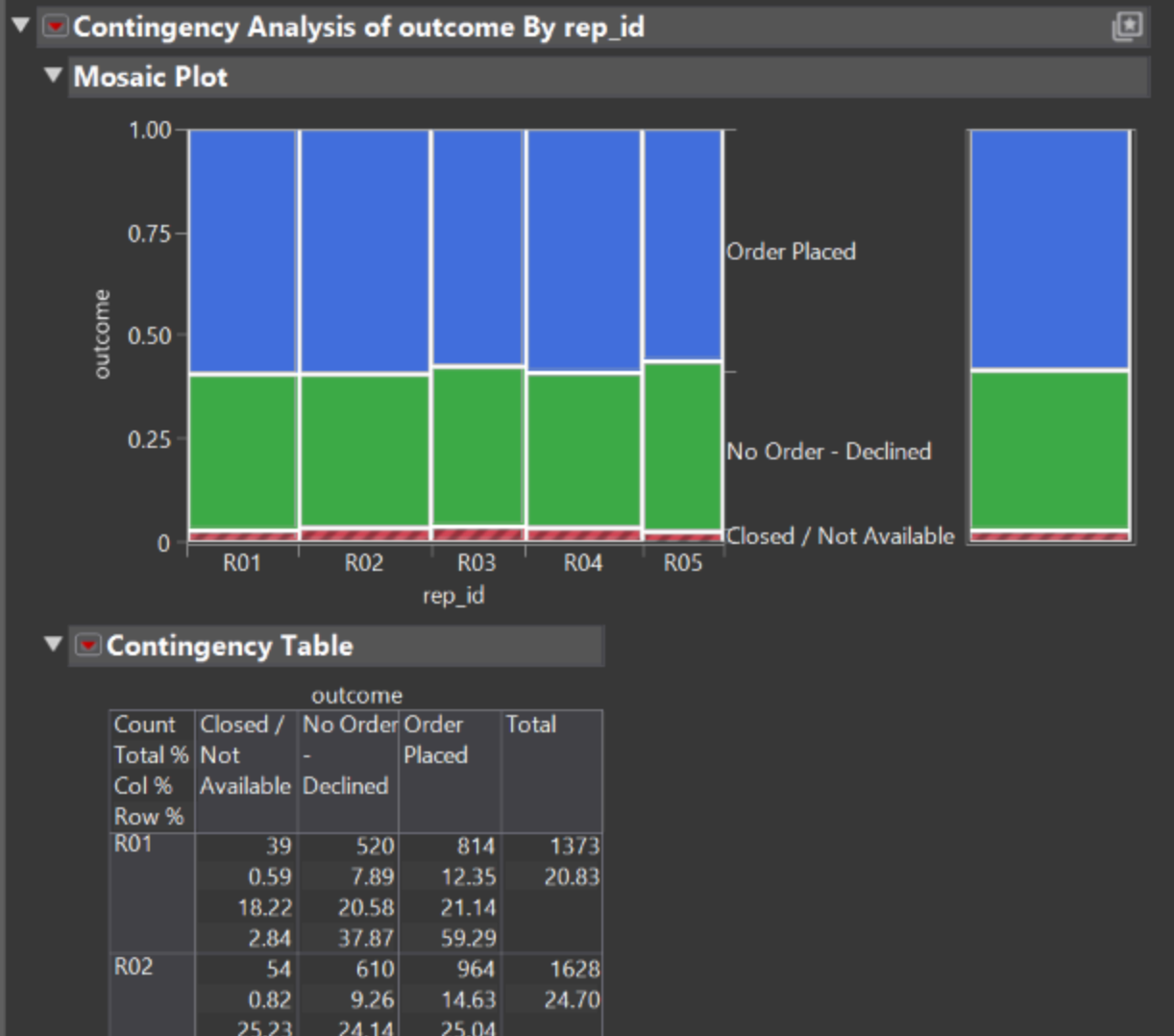
Source	DF	Sum of Squares	Mean Square	F Ratio	Prob > F
sku	9	2750.2	305.580	0.9993	0.4379
Error	7874	2407707.4	305.779		
C. Total	7883	2410457.7			

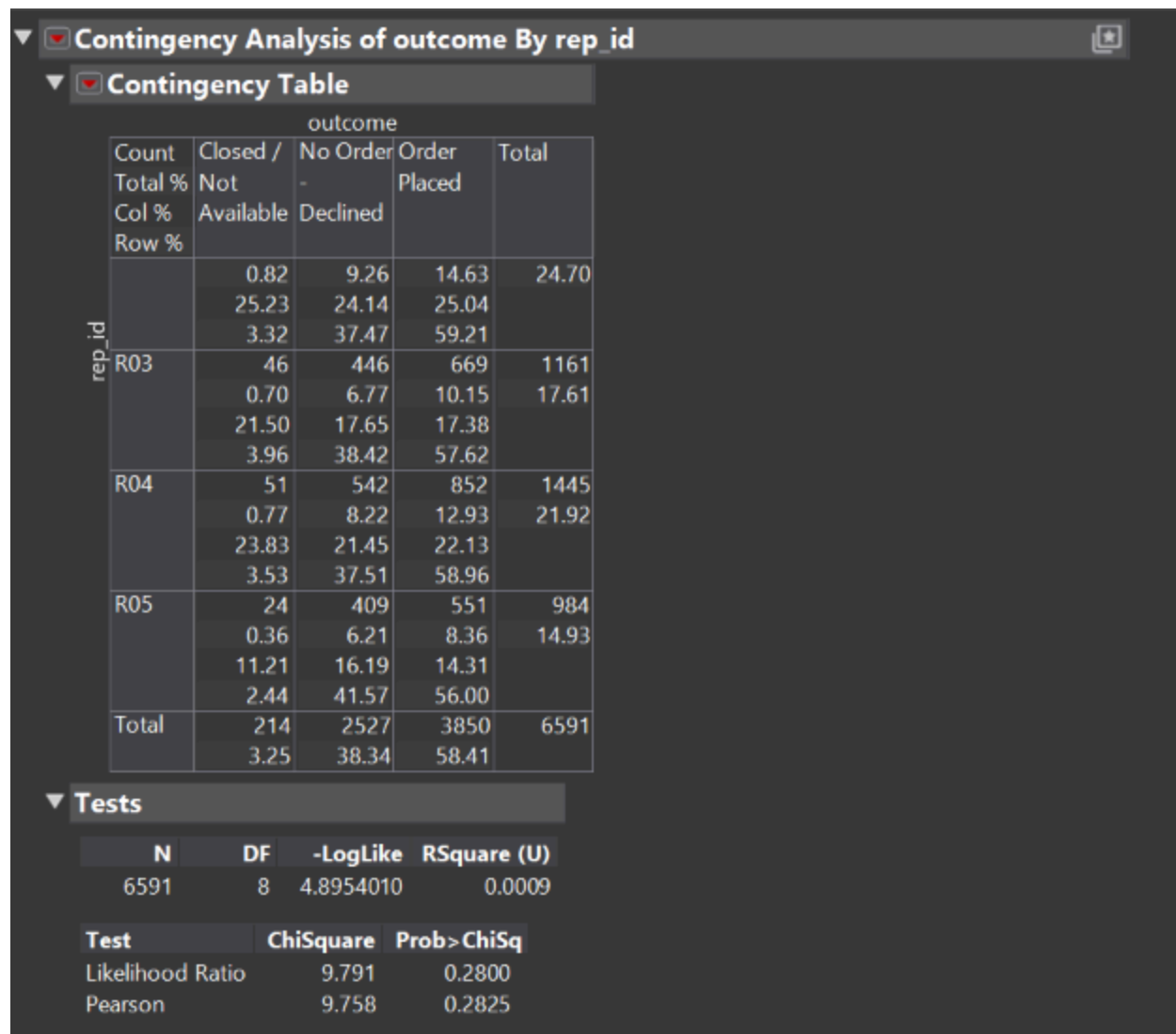
▼ Means for Oneway Anova

Level	Number	Mean	Std Error	Lower 95%	Upper 95%
YG-DRK-250	769	18.6788	0.63058	17.443	19.915
YG-DRK-500	776	18.4961	0.62773	17.266	19.727
YG-GRK-450	771	17.9948	0.62976	16.760	19.229
YG-KID-100	741	17.1080	0.64238	15.849	18.367
YG-MNG-500	837	17.3142	0.60442	16.129	18.499
YG-PLN-500	833	17.4766	0.60587	16.289	18.664
YG-PLN-1000	784	17.9439	0.62452	16.720	19.168
YG-PSN-500	756	16.9180	0.63598	15.671	18.165
YG-STRW-500	810	18.5975	0.61441	17.393	19.802
YG-VNL-250	807	17.7831	0.61556	16.576	18.990

Std Error uses a pooled estimate of error variance

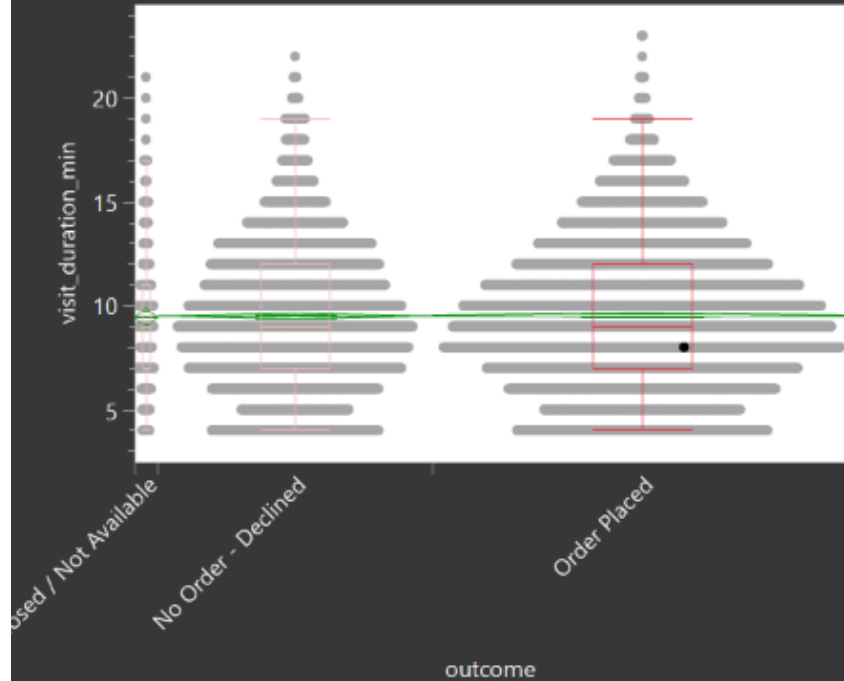
EDA_Bivariate_Conversion_by_Rep.png





EDA_Bivariate_Duration_by_Outcome.png

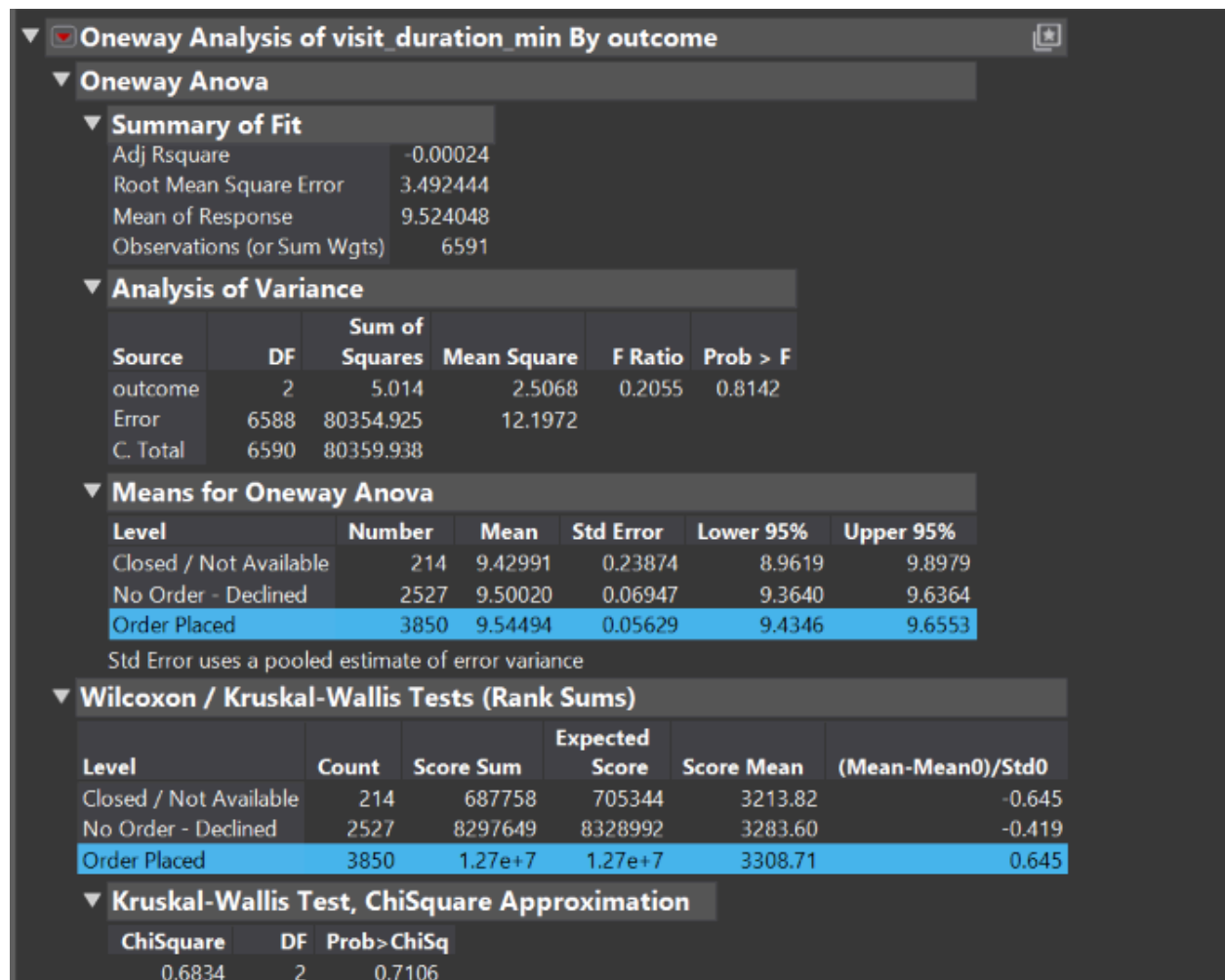
▼ **Oneway Analysis of visit_duration_min By outcome**



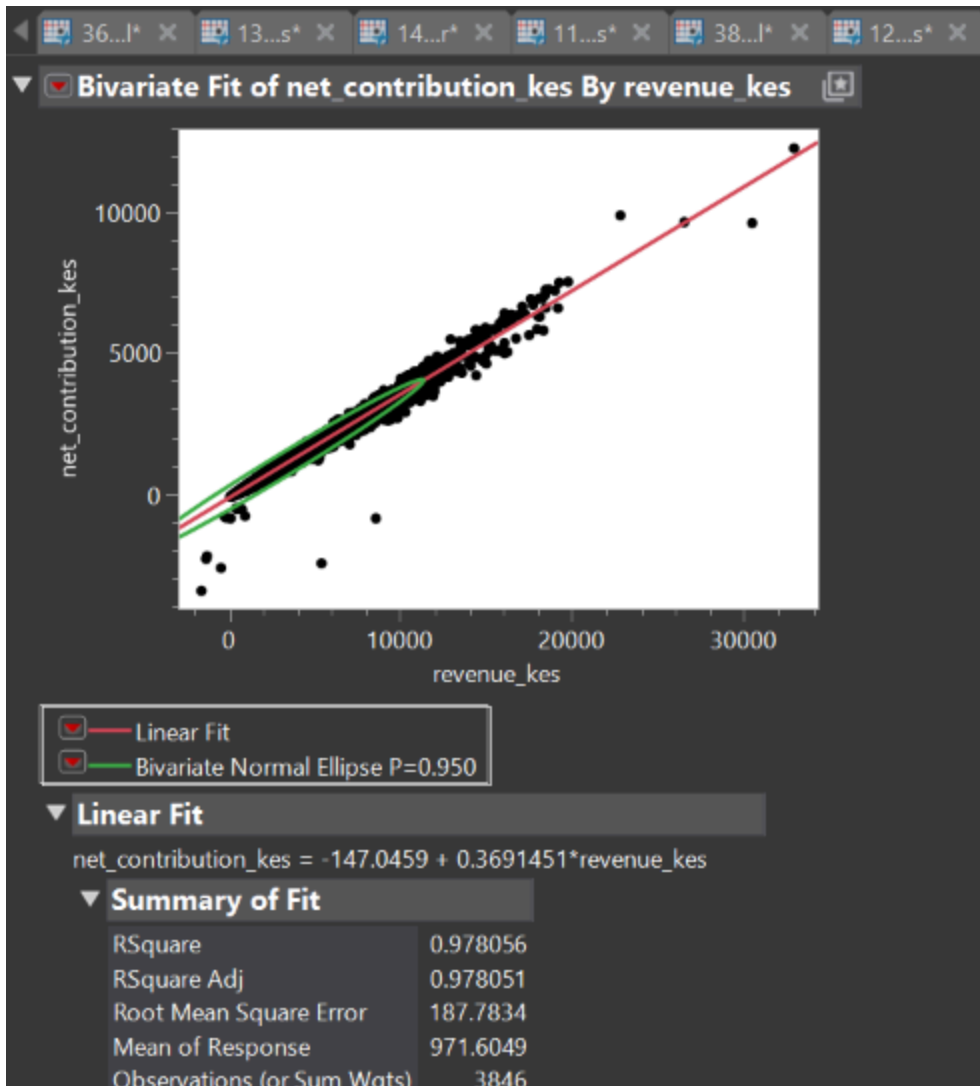
▼ **Oneway Anova**

▼ **Summary of Fit**

Rsquare	6.239e-5
Adj Rsquare	-0.00024
Root Mean Square Error	3.492444
Mean of Response	9.524048
Observations (or Sum Wgts)	6591



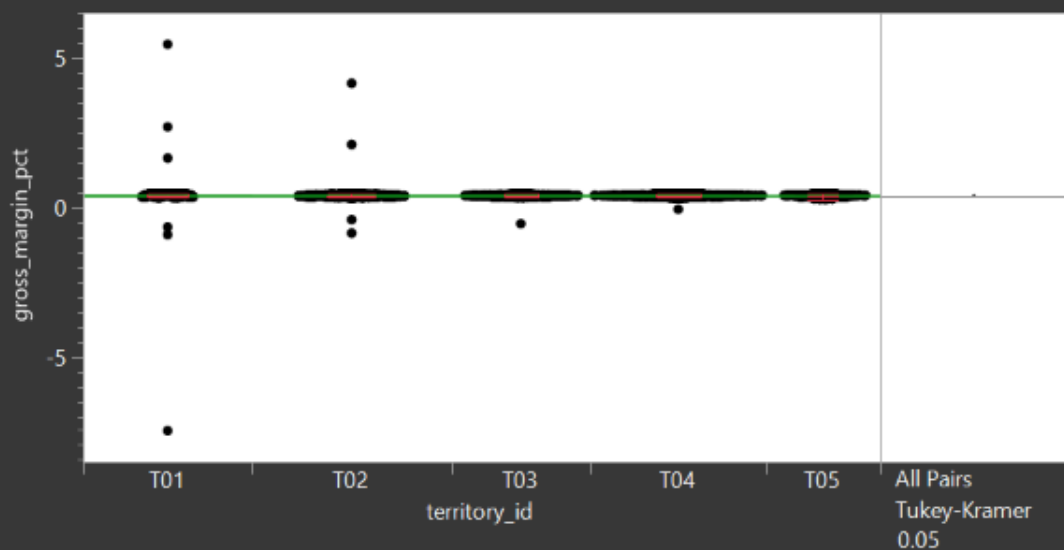
EDA_Bivariate_NetContribution_vs_Revenue.png



</

EDA_Bivariate_GrossMarginPct_by_Territory.png

▼ **Oneway Analysis of gross_margin_pct By territory_id**



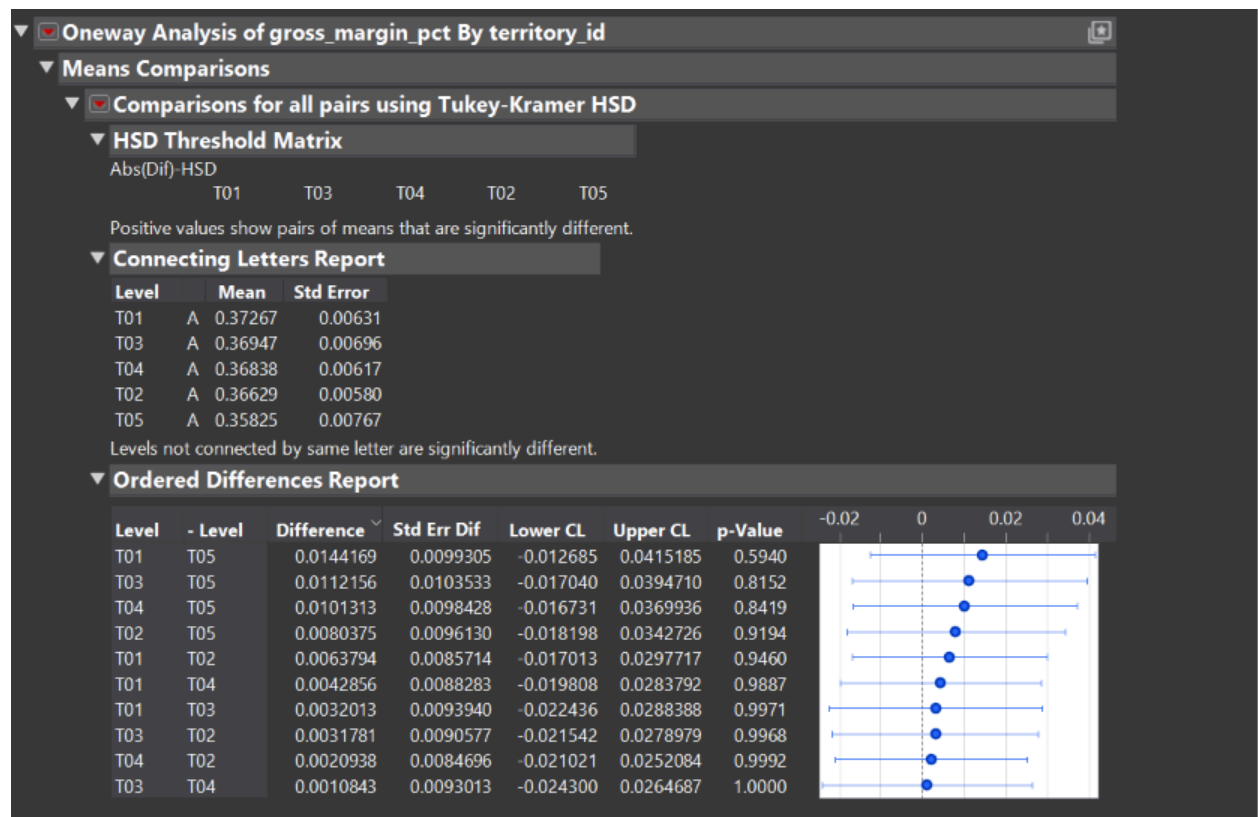
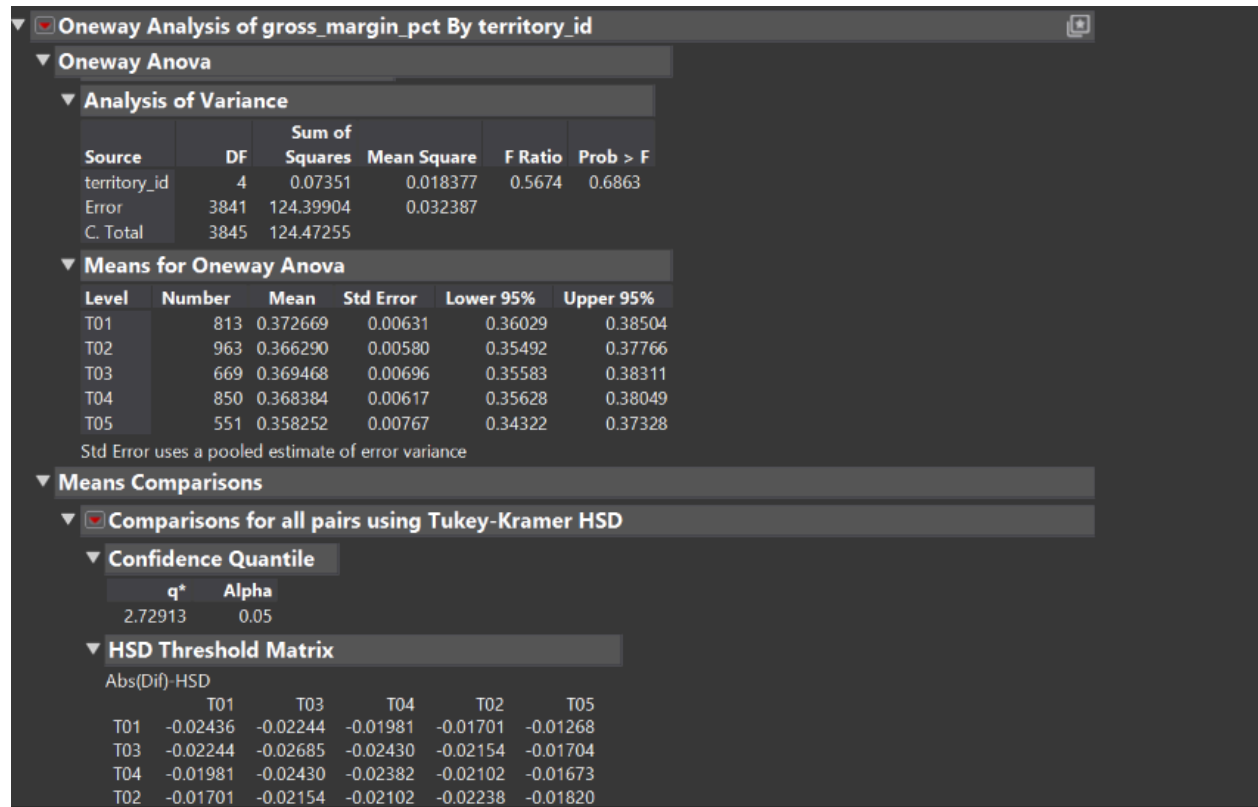
▼ **Oneway Anova**

▼ **Summary of Fit**

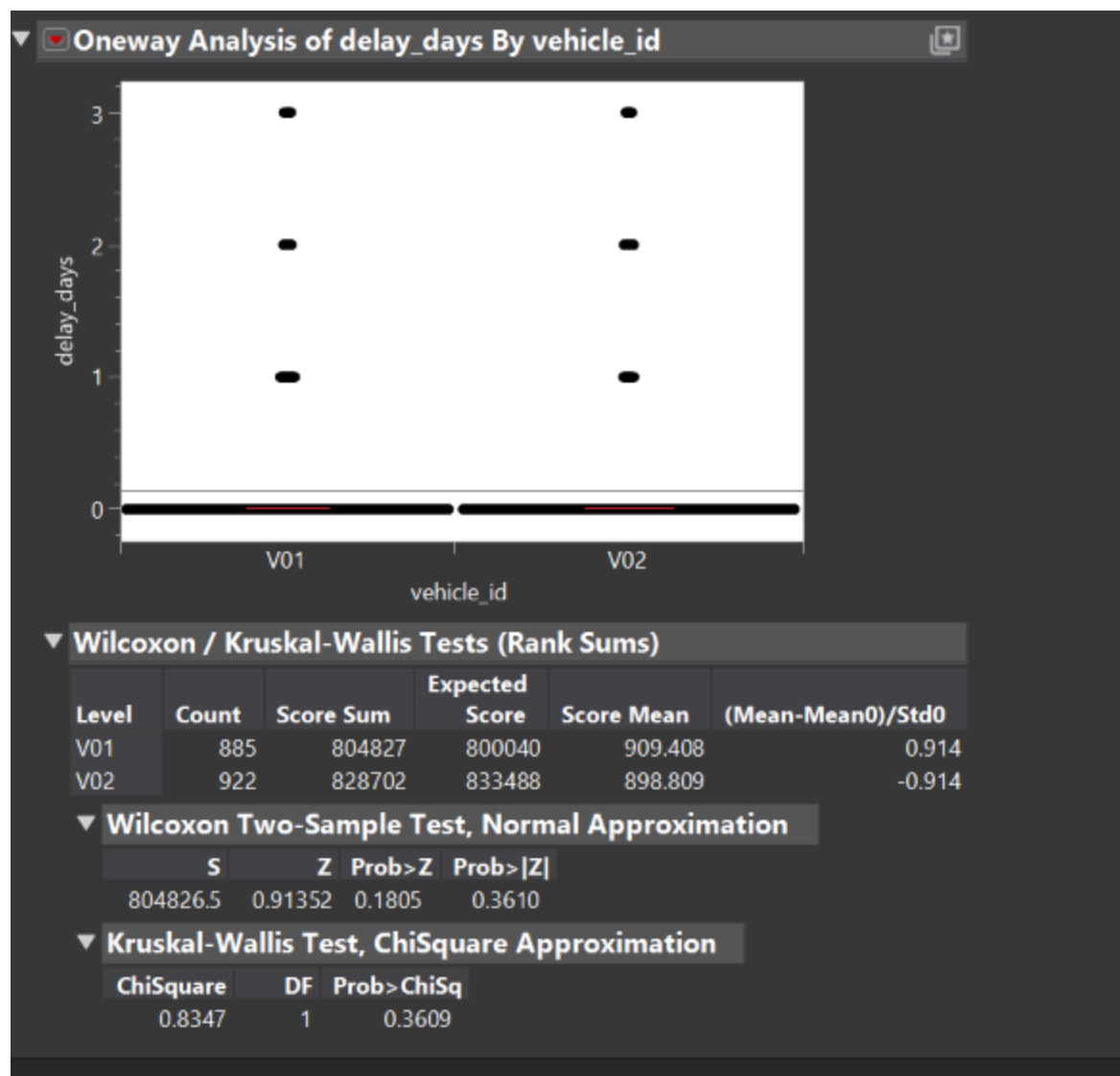
Rsquare	0.000591
Adj Rsquare	-0.00045
Root Mean Square Error	0.179964
Mean of Response	0.367502
Observations (or Sum Wgts)	3846

▼ **Analysis of Variance**

Source	DF	Sum of Squares	Mean Square	F Ratio	Prob > F
territory_id	4	0.07351	0.018377	0.5674	0.6863



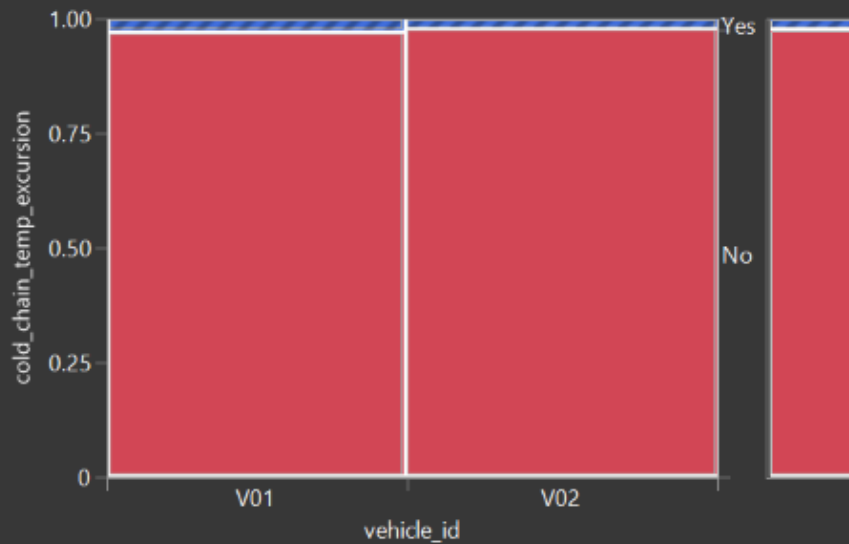
EDA_Bivariate_DelayDays_by_Vehicle.png



EDA_Bivariate_TempExcursion_by_Vehicle.png

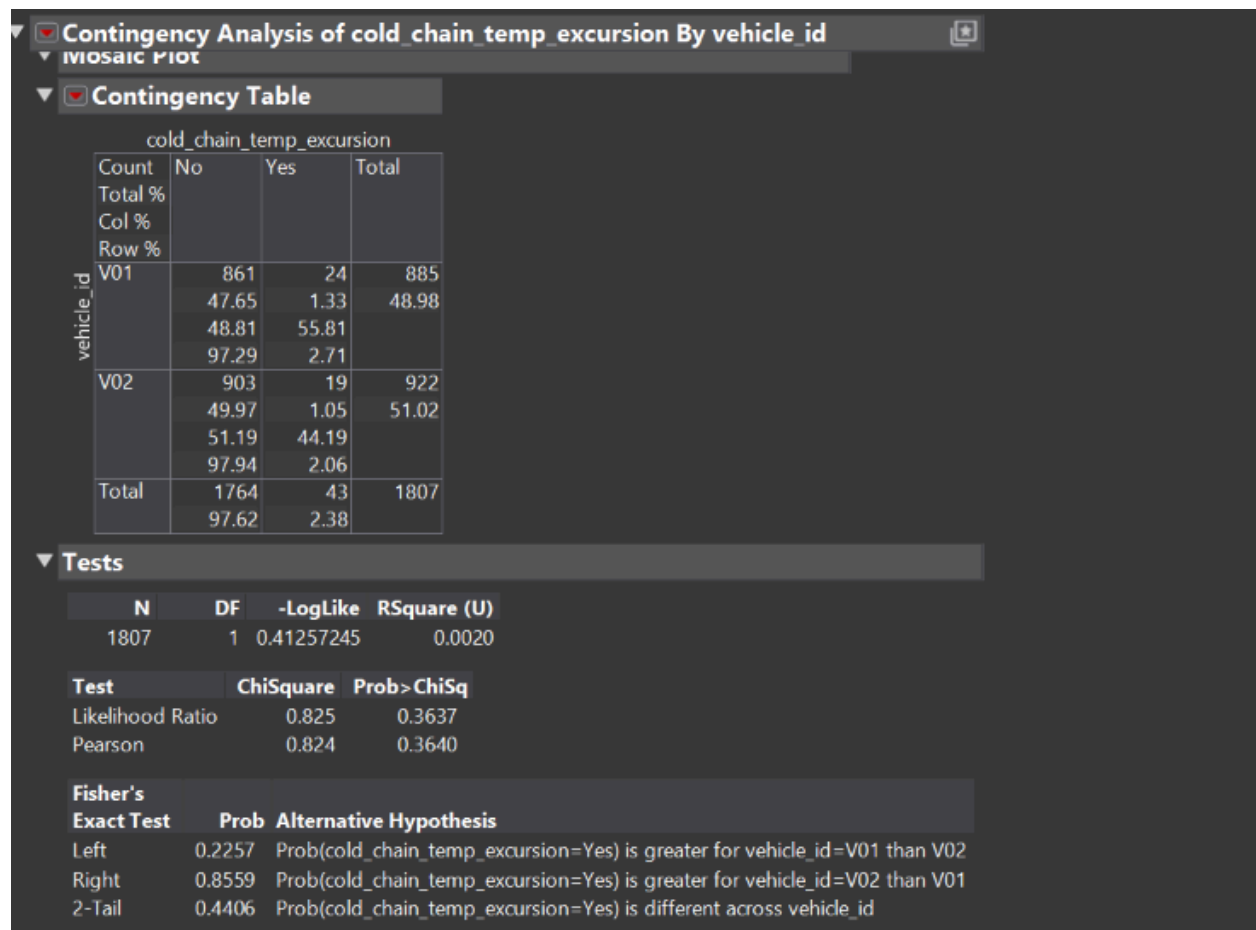
Contingency Analysis of cold_chain_temp_excursion By vehicle_id

Mosaic Plot



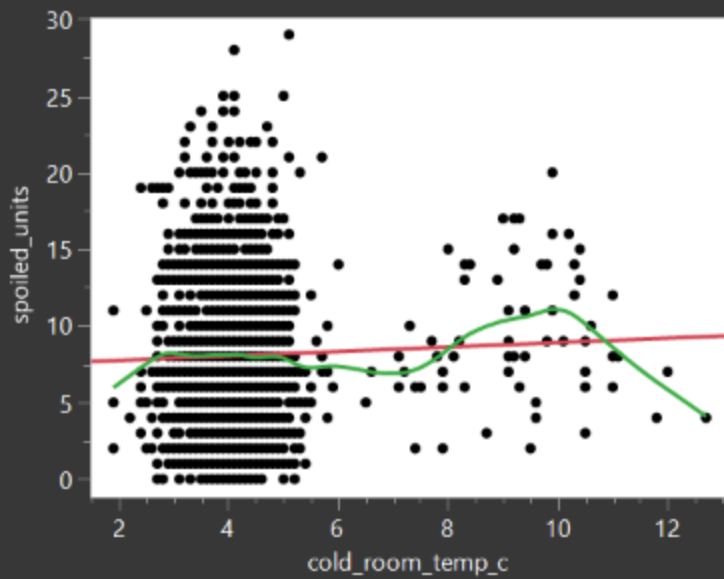
Contingency Table

		cold_chain_temp_excursion		
vehicle_id	Count	No	Yes	Total
	Total %			
	Col %			
	Row %			
	V01	861	24	885
		47.65	1.33	48.98
V02		48.81	55.81	
		97.29	2.71	
	V02	903	19	922
		48.07	1.05	51.02



EDA_Bivariate_Spoilage_vs_Temp.png

▼ ☒ Bivariate Fit of spoiled_units By cold_room_temp_c



- ☒ Linear Fit
- ☒ Smoothing Spline Fit, lambda=1

▼ **Linear Fit**

$$\text{spoiled_units} = 7.4567082 + 0.1450085 \cdot \text{cold_room_temp_c}$$

▼ **Summary of Fit**

RSquare	0.001264
RSquare Adj	0.000712
Root Mean Square Error	4.914484
Mean of Response	8.069061
Observations (or Sum Wgts)	1810

▼ ☒ **Bivariate Fit of spoiled_units By cold_room_temp_c** 

▼ **Linear Fit**

▼ **Summary of Fit**

Root Mean Square Error	4.914484
Mean of Response	8.069061
Observations (or Sum Wgts)	1810

▼ **Lack Of Fit**

Source	DF	Sum of Squares	Mean Square	F Ratio
Lack Of Fit	75	1832.632	24.4351	1.0122
Pure Error	1733	41834.452	24.1399	Prob > F
Total Error	1808	43667.084		0.4507

Max RSq
0.0432

▼ **Analysis of Variance**

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	1	55.283	55.2832	2.2890
Error	1808	43667.084	24.1521	Prob > F
C. Total	1809	43722.367		0.1305

▼ **Parameter Estimates**

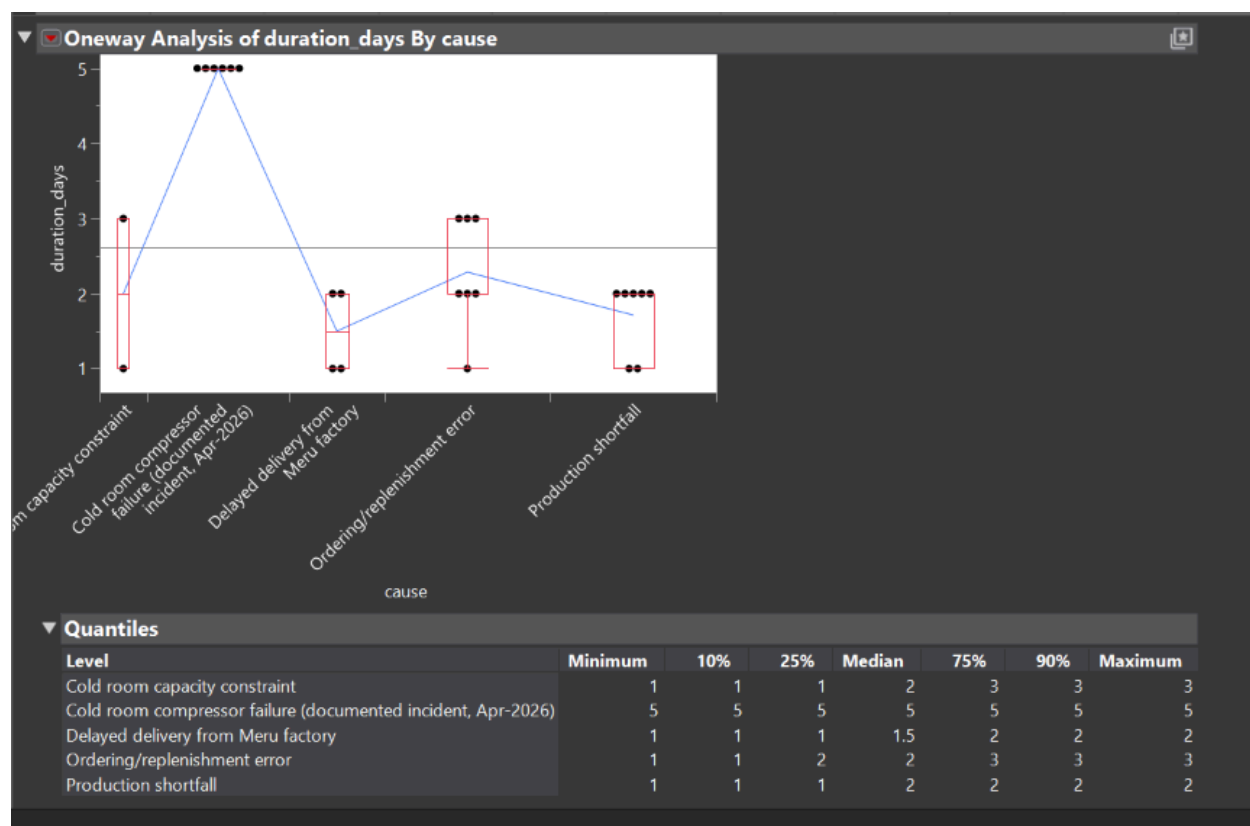
Term	Estimate	Std Error	t Ratio	Prob> t
Intercept	7.4567082	0.420908	17.72	<.0001*
cold_room_temp_c	0.1450085	0.095846	1.51	0.1305

▼ **Smoothing Spline Fit, lambda=1**

R-Square	0.009943
Sum of Squares Error	43287.64

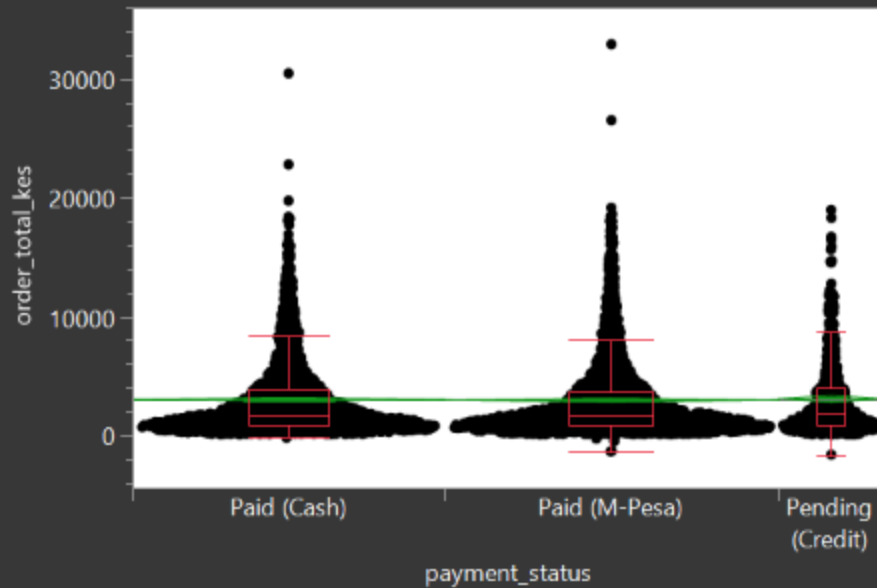
Change Lambda: 

EDA_Bivariate_StockoutDuration_by_Cause.png



EDA_Bivariate_OrderTotal_by_PaymentStatus.png

▼ **Oneway Analysis of order_total_kes By payment_status**



▼ **Quantiles**

Level	Minimum	10%	25%	Median	75%	90%	Maximum
Paid (Cash)	-170	441.6	864.9	1768.5	3905.2	7884.16	30470.4
Paid (M-Pesa)	-1330	420	802.5	1680	3788.1	7527.36	32910
Pending (Credit)	-1600	432	816.8	1820	4020.925	7949.3	18980

▼ **Oneway Anova**

▼ **Summary of Fit**

Rsquare	0.000107
Adj Rsquare	-0.00041
Root Mean Square Error	3396.411
Mean of Response	3030.382

